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Question # 1 of 10 (Start time: 03:39:57 PM, 28 June 2022)

An in-place sorting algorithm is one that _____ use(s) additional array for storage.

Select the correct option

☐	Permanently
☐	Always
☐	Sometime
☒	Does not

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Question # 2 of 10 (Start time: 03:40:40 PM, 28 June 2022)

Developing a dynamic programming algorithm generally involves _____ separate steps.

Select the correct option

☐	Four
☐	One
☐	Three
☒	Two

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Question # 3 of 10 (Start time: 03:40:57 PM, 28 June 2022)

Merge sort is a/an _____ and _____ sorting algorithm.

Select the correct option

☐	In-place, not stable one
☐	In-place, stable one
☐	Not in-place, not stable one
☒	Not in-place, stable one

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Question # 5 of 10 (Start time: 03:44:01 PM, 28 June 2022)

Selection sort is a _____ sorting algorithm.

Select the correct option

☐	Stable
☐	Not In-Place
☐	in-partition
☒	In-place

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Question # 6 of 10 (Start time: 03:44:34 PM, 28 June 2022)

Chain matrix multiplication problem can be solved through _____ strategy.

Select the correct option

☐	greedy
☐	divide and conquer
☐	Implementing arrays
☒	dynamic programming

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Question # 7 of 10 (Start time: 03:46:04 PM, 28 June 2022)

The only way to convert an empty string into a string of j characters is by doing j insertions, represented as _____.

Select the correct option

☐	$E(1, j) = j$
☒	$E(0, j) = j$
☐	$E(i, j) = 1$
☐	$E(i, 0) = i$

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Question # 8 of 10 (Start time: 03:46:40 PM, 28 June 2022)

After partitioning array in Quick sort, pivot is placed in a position such that

Select the correct option

☐	Pivot is the first element of array
☐	Pivot is the last element of array
☐	Values larger than pivot are on left and smaller than pivot are on right
☒	Values smaller than pivot are on left and larger than pivot are on right

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Question # 9 of 10 (Start time: 03:47:41 PM, 28 June 2022)

As per algorithm of dynamic programing, we need to store the result(s) of _____

Select the correct option

☐	First sub-problem only
☐	Final solution only
☐	Best solution only
☒	Intermediate sub-problems

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Question # 1 of 10 (Start time: 03:50:37 PM, 28 June 2022)

In chain matrix multiplication, if there are n items, there are _____ ways in which outer most pair of parentheses can placed.

Select the correct option

☐	n^2
☒	$n-1$
☐	$n+1$
☐	$2n$

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Question # 2 of 10 (Start time: 03:50:59 PM, 28 June 2022)

Matrix multiplication is a(n) _____ operation.

Select the correct option

☐	Neither commutative nor associative
☒	Associative
☐	Transitive
☐	Commutative

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Question # 3 of 10 (Start time: 03:51:32 PM, 28 June 2022)

We can use the optimal substructure property to devise a _____ formulation of the edit distance problem.

Select the correct option

☐	Iterative
☐	Selective
☒	Recursive
☐	Optimum

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Question # 4 of 10 (Start time: 03:51:50 PM, 28 June 2022)

Which one of the following problems can be solved using dynamic problem?

Select the correct option

☒	Matrix chain multiplication problem
☐	Fractional knapsack problem
☐	Greedy search problem
☐	Bubble sort problem

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Question # 6 of 10 (Start time: 03:54:19 PM, 28 June 2022)

For comparison-based sorting algorithms, it is _____ possible to sort more efficiently than $\Omega(n \log n)$ time.

Select the correct option

☐	Sometimes not
☒	Not
☐	Always
☐	Sometimes

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Question # 8 of 10 (Start time: 03:57:15 PM, 28 June 2022)

Edit distance algorithm based on _____ strategy.

Select the correct option

☐	Greedy
☐	Divide and Conquer
☒	Dynamic Programming
☐	Brute force

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Question # 9 of 10 (Start time: 03:58:15 PM, 28 June 2022)

_____ belongs to Dynamic programming.

Select the correct option

☐	Divide and conquer
☒	Edit distance
☐	Merge sort
☐	Heap sort

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Question # 1 of 10 (Start time: 04:01:09 PM, 28 June 2022)

In quick sort algorithm, pivots form a _____.

Select the correct option

☐	Graph
☐	Stack
☐	Queue
☒	Binary Search Tree

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Question # 2 of 10 (Start time: 04:01:26 PM, 28 June 2022)

Dynamic Programming approach is usually useful in solving _____ problems.

Select the correct option

☐	loop
☐	normal
☒	optimization
☐	array

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Question # 3 of 10 (Start time: 04:02:08 PM, 28 June 2022)

We can improve the performance of quick sort if we could be able to _____.

Select the correct option

☐	Eliminate recursive calls
☐	Skip input elements somehow
☒	Select two or more pivots
☐	Skip any sub-array completely

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Question # 4 of 10 (Start time: 04:03:18 PM, 28 June 2022)

Which one of the following sorting algorithms is the fastest?

Select the correct option

☐	Heap sort
☐	Insertion sort
☒	Quick sort
☐	Merge sort

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Question # 6 of 10 (Start time: 04:04:34 PM, 28 June 2022)

In chain matrix multiplication, if there are n items, there are _____ ways in which outer most pair of parentheses can placed.

Select the correct option

☒	$n-1$
☐	$2n$
☐	n^2
☐	$n+1$

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Question # 8 of 10 (Start time: 04:06:01 PM, 28 June 2022)

An in-place sorting algorithm is one that _____ use(s) additional array for storage.

Select the correct option

☐	Permanently
☐	Always
☒	Does not
☐	Sometime

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Question # 9 of 10 (Start time: 04:06:16 PM, 28 June 2022)

Edit distance algorithm based on _____ strategy.

Select the correct option

☒	Dynamic Programming
☐	Brute force
☐	Greedy
☐	Divide and Conquer

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Question # 2 of 10 (Start time: 04:16:27 PM, 28 June 2022)

The only way to convert an empty string into a string of j characters is by doing j insertions, represented as _____.

Select the correct option

☐	$E(i, j) = i$
☐	$E(1, j) = j$
☐	$E(i, 0) = i$
☒	$E(0, j) = j$

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Question # 3 of 10 (Start time: 04:16:45 PM, 28 June 2022)

Radix sort performs sorting the numbers _____ digit(s) at a time.

Select the correct option

☒	One
☐	Three
☐	Two
☐	All

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Question # 4 of 10 (Start time: 04:17:45 PM, 28 June 2022)

The formula for calculating the Catalan number is _____.

Select the correct option

- | | |
|----------------------------------|--------------------------------------|
| ☐ | $C(n) = \frac{1}{n-1} \binom{2n}{n}$ |
| ☒ | $C(n) = \frac{1}{n+1} \binom{2n}{n}$ |
| ☐ | $C(n) = \frac{1}{n-1} \binom{n}{2n}$ |
| ☐ | $C(n) = \frac{1}{n+1} \binom{n}{2n}$ |

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Question # 5 of 10 (Start time: 04:18:11 PM, 28 June 2022)

Edit distance algorithm based on _____ strategy.

Select the correct option

☐	Divide and Conquer
☒	Dynamic Programming
☐	Brute force
☐	Greedy

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Question # 6 of 10 (Start time: 04:18:18 PM, 28 June 2022)

For average-case time analysis of quick sort algorithm, pivot selection is on average basis from _____.

Select the correct option

☐	values greater than 5
☐	all values except even
☐	half of the input values
☒	all possible random values

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Question # 7 of 10 (Start time: 04:19:02 PM, 28 June 2022)

In quick sort algorithm, _____ decides nature of Binary Search Tree formed by pivots.

Select the correct option

☐	middle element from input
☐	largest element from input
☐	smallest element from input
☒	rank of the pivot

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Question # 1 of 10 (Start time: 10:10:13 AM, 08 June 2022)

Brute-force algorithm for 2D-Maxima is operated by comparing _____ pairs of points.

Select the correct option

☐	Some
☒	All
☐	Most
☐	Two

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Question # 2 of 10 (Start time: 10:11:55 AM, 08 June 2022)

In the following loop statement:

for i ← 1 to n

For loop executes _____ times.

Select the correct option

☐	n^2
☒	n
☐	$\log n$
☐	It is not correct statement

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Question # 3 of 10 (Start time: 10:13:12 AM, 08 June 2022)

Two functions are said to be asymptotically equivalent if they have _____.

Select the correct option

☐	same input
☒	same growth rate for large n
☐	same number of polynomials
☐	been proved as equivalent

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Question # 5 of 10 (Start time: 10:16:17 AM, 08 June 2022)

The O-notation is used to state only the asymptotic _____ bounds.

Select the correct option

☐	Lower
☒	Upper
☐	Two
☐	Both lower & upper

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Question # 7 of 10 (Start time: 10:18:23 AM, 08 June 2022)

Asymptotic notations are used to describe _____ of an algorithm.

Select the correct option

☐	compile time
☐	size
☒	running time
☐	length

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Question # 8 of 10 (Start time: 10:19:40 AM, 08 June 2022)

Efficient algorithm requires less computational _____.

Select the correct option

☐	running time
☒	memory and running time
☐	memory
☐	notation

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Question # 9 of 10 (Start time: 10:20:04 AM, 08 June 2022)

In Ram, each instruction involves performing some basic operation on how many values in the machines memory?

Select the correct option

☐	Three
☐	Four
☒	Two
☐	One

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Question # 10 of 10 (Start time: 10:21:26 AM, 08 June 2022)

Random Access Machine (RAM) can execute _____ instructions.

Select the correct option

☒	Logical and arithmetic
☐	Parallel
☐	Only arithmetic
☐	Only logical

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Question # 1 of 10 (Start time: 10:39:16 AM, 08 June 2022)

If the time complexity of an algorithm is $O(n)$, then it is called _____ time complexity.

Select the correct option

☐	Exponential
☐	Average
☐	Constant
☒	Linear

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Question # 2 of 10 (Start time: 10:40:47 AM, 08 June 2022)

For 2D Maxima problem, Plane Sweep algorithm first of all _____.

Select the correct option

☐	Pushes all points on stack
☐	Output the elements
☐	Delete some points
☒	Sorts all points

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Question # 3 of 10 (Start time: 10:42:01 AM, 08 June 2022)

In Ram, each instruction involves performing some basic operation on how many values in the machines memory?

Select the correct option

☐	One
☐	Three
☒	Two
☐	Four

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Question # 4 of 10 (Start time: 10:42:10 AM, 08 June 2022)

if A dominates B, and B dominates C, then it is necessary that

Select the correct option

☒	C dominates A
☐	B dominates A
☐	C dominates B
☐	A dominates C

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Question # 5 of 10 (Start time: 10:43:25 AM, 08 June 2022)

Approach of solving geometric problems by sweeping a line across the plane is called _____ sweep.

Select the correct option

☐	Line
☒	Plane
☐	Box
☐	Cube

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Question # 6 of 10 (Start time: 10:43:50 AM, 08 June 2022)

Cubic function will _____ a quadratic function.

Select the correct option

☐	prove
☐	find
☒	overtake
☐	be equal to

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Question # 7 of 10 (Start time: 10:44:10 AM, 08 June 2022)

While analysis of the brute-force maxima algorithm, an array sorted in the reverse order is the type of _____ case input.

Select the correct option

☒	Worst
☐	Best
☐	Somewhat bad
☐	Average

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Question # 8 of 10 (Start time: 10:44:56 AM, 08 June 2022)

Dominance relationship has _____ property.

Select the correct option

☐	Symmetric,
☐	Reflexive
☐	Substitution
☒	Transitive

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Question # 9 of 10 (Start time: 10:46:12 AM, 08 June 2022)

In Random Access Machine, Instructions are executed

Select the correct option

☐	in parallel
☒	one by one
☐	in the form of bits
☐	in the form of bytes

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Question # 10 of 10 (Start time: 10:46:34 AM, 08 June 2022)

For _____ values of n, any algorithm is fast enough.

Select the correct option

☐	Medium
☐	Infinity
☒	Small
☐	Large

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Question # 1 of 10 (Start time: 10:50:30 AM, 08 June 2022)

Two functions are said to be asymptotically equivalent if they have _____.

Select the correct option

☐	been proved as equivalent
☐	same number of polynomials
☒	same growth rate for large n
☐	same input

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Question # 2 of 10 (Start time: 10:50:49 AM, 08 June 2022)

There are no hard formal rules to the syntax of the _____ code.

Select the correct option

☐	Programming
☐	Basic
☒	Pseudo
☐	Assembly

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Question # 3 of 10 (Start time: 10:52:22 AM, 08 June 2022)

Which of the following is calculated with Big Omega notation?

Select the correct option

☐	Medium bounds
☒	Lower bounds
☐	Upper bounds
☐	Both upper and lower bounds

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اسلام علیکم:

امید کرتا ہوں سب خیریت سے ہوں گئے۔ آنے والے تمام نیو طالب علم کو میں ورچوئل یونیورسٹی آف پاکستان میں خوش آمدید کرتا ہوں۔
اگر کسی سٹوڈنٹ کو کسی بھی قسم کی ہیلپ چاہے ہو تو ہم سے رابطہ کر سکتا ہے جیسا کہ: پاسٹ پیپرز، کورس سلیکشن، پروجیکٹ سلیکشن، ڈیٹ شیٹ،
فیس پے اور چینج سٹڈی پروگرام وغیرہ۔

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VU LMS

اگر کوئی سٹوڈنٹ کسی مجبوری کی وجہ سے یا جاب ہو لڈز ہونے کی وجہ سے اپنا ایل ایم ایس خود ہینڈل نہیں کر سکتا تو وہ ہم سے پیڈ سروس حاصل کر
سکتا ہے کوشش کرے کے آپ ہر چیز خود اٹیمپٹ کریں اگر نہیں کر سکتے تو سروس حاصل کریں۔

Email:

waqasahmeduog@gmail.com

انشاء اللہ آپ ہماری سروس سے مطمئن ہوں گے۔

Whatsapp: 03066295623 , 03404643855

مزید اگر کسی رہنمائی کی ضرورت ہو تو وہ رابطہ کر سکتا ہے۔

Question # 4 of 10 (Start time: 10:53:46 AM, 08 June 2022)

Asymptotic notations are used to describe _____ of an algorithm.

Select the correct option

☒	running time
☐	compile time
☐	length
☐	size

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Question # 5 of 10 (Start time: 10:54:46 AM, 08 June 2022)

Brute-force algorithm for 2D-Maxima is operated by comparing _____ pairs of points.

Select the correct option

☐	Two
☒	All
☐	Most
☐	Some

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Question # 6 of 10 (Start time: 10:55:11 AM, 08 June 2022)

The brute-force algorithm for 2D-Maxima runs in order $O(___)$ time.

Select the correct option

☒	$n \cdot n$
☐	n
☐	$n(\log n)$
☐	n^3

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Question # 7 of 10 (Start time: 10:56:48 AM, 08 June 2022)

In addition to passing in the array itself to merge sort algorithm, we will pass in _____ other arguments which are indices.

Select the correct option

☐	5
☐	3
☐	4
☒	2

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Question # 8 of 10 (Start time: 10:57:58 AM, 08 June 2022)

Execution time of an algorithm can be measured by _____.

Select the correct option

☐	Brute force approach
☒	Mathematical analysis
☐	both brute force and divide and conquer approach
☐	Divide and conquer approach

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Question # 8 of 10 (Start time: 10:57:58 AM, 08 June 2022)

Execution time of an algorithm can be measured by _____.

Select the correct option

☐	Brute force approach
☒	Mathematical analysis
☐	both brute force and divide and conquer approach
☐	Divide and conquer approach

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Question # 9 of 10 (Start time: 10:59:09 AM, 08 June 2022)

The main purpose of mathematical analysis is measuring the _____ required by the algorithm.

Select the correct option

☐	Space
☒	Execution time and memory
☐	Execution time
☐	Inputs & outputs

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Question # 10 of 10 (Start time: 11:00:21 AM, 08 June 2022)

In asymptotical analysis of $n^*(5 + 2) - 3$, as n becomes large, the dominant (fastest growing) term is some constant times _____

Select the correct option

☒	$n*n$
☐	n_1
☐	n
☐	$n+1$

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Question # 1 of 10 (Start time: 11:02:32 AM, 08 June 2022)

In Running Time Analysis, we are also concerned about the _____ required by the algorithm.

Select the correct option

☐	Space
☐	code
☐	Memory
☒	Both space and memory

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Question # 3 of 10 (Start time: 11:05:05 AM, 08 June 2022)

Algorithm is a sequence of computational steps that _____ the input into output.

Select the correct option

☐	integrate
☐	assign
☐	merge
☒	transform

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Question # 4 of 10 (Start time: 11:05:49 AM, 08 June 2022)

In asymptotical analysis of $n^*(5 + 2) - 3$, as n becomes large, the dominant (fastest growing) term is some constant times _____

Select the correct option

☐	n
☒	$n*n$
☐	n_1
☐	$n+1$

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Question # 5 of 10 (Start time: 11:06:08 AM, 08 June 2022)

The main purpose of mathematical analysis is measuring the _____ required by the algorithm.

Select the correct option

☐	Space
☒	Execution time and memory
☐	Execution time
☐	Inputs & outputs

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Question # 7 of 10 (Start time: 11:07:49 AM, 08 June 2022)

In plane sweep approach of solving geometric problems, a _____ is swept across the plane.

Select the correct option

☐	Plane
☐	Line
☐	Cube
☐	Box

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Question # 8 of 10 (Start time: 11:09:36 AM, 08 June 2022)

There are no hard formal rules to the syntax of the _____ code.

Select the correct option

☐	Basic
☐	Assembly
☒	Pseudo
☐	Programming

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Question # 9 of 10 (Start time: 11:09:52 AM, 08 June 2022)

Using _____ algorithm, efficiency is not given much importance.

Select the correct option

☐	Merge sort
☐	Brute Force
☐	Greedy
☒	Processing

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Question # 10 of 10 (Start time: 11:11:04 AM, 08 June 2022)

Which of the following is calculated with Big Omega notation?

Select the correct option

☐	Both upper and lower bounds
☐	Upper bounds
☐	Medium bounds
☒	Lower bounds

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Question # 1 of 10 (Start time: 11:12:53 AM, 08 June 2022)

Two functions are said to be asymptotically equivalent if they have _____.

Select the correct option

☒	same growth rate for large n
☐	been proved as equivalent
☐	same input
☐	same number of polynomials

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Question # 2 of 10 (Start time: 11:13:00 AM, 08 June 2022)

Algorithm is a sequence of computational steps that _____ the input into output.

Select the correct option

☐	merge
☐	assign
☒	transform
☐	integrate

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Question # 3 of 10 (Start time: 11:13:08 AM, 08 June 2022)

If the indices passed to merge sort algorithm are _____, then this means that there is only one element to sort.

Select the correct option

☒	Equal
☐	Not Equal
☐	Small
☐	Large

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Question # 7 of 10 (Start time: 11:18:41 AM, 08 June 2022)

In Random Access Machine, Instructions are executed

Select the correct option

☐	in the form of bits
☐	in parallel
☒	one by one
☐	in the form of bytes

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Question # 8 of 10 (Start time: 11:18:47 AM, 08 June 2022)

_____ of reference is an important fact of current processor technology.

Select the correct option

☐	Assigning
☒	Locality
☐	Defining
☐	Formality

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Question # 9 of 10 (Start time: 11:19:08 AM, 08 June 2022)

In average-case time the probability of seeing input is denoted by_____.

Select the correct option

☐	$p[l]$
☐	$p<l>$
☐	$p\{l\}$
☒	$p(l)$

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Question # 10 of 10 (Start time: 11:19:53 AM, 08 June 2022)

The _____ is the weighted sum of running times with weights.

Select the correct option

☐	both best-case and average-case.
☒	average-case time
☐	best-case time
☐	worse-case time

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Question # 1 of 10 (Start time: 08:20:19 PM, 08 June 2022)

In asymptotical analysis of $n(n - 3)$ and $4n^*n$, as n becomes large, the dominant (fastest growing) term is some constant times_____.

Select the correct option

☐	$n+1$
☐	n
☐	$n-1$
☒	n^*n

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VU KMS

اسلام علیکم:

امید کرتا ہوں سب خیریت سے ہوں گئے۔ آنے والے تمام نیو طالب علم کو میں ورچوئل یونیورسٹی آف پاکستان میں خوش آمدید کرتا ہوں۔
اگر کسی سٹوڈنٹ کو کسی بھی قسم کی ہیلپ چاہے ہو تو ہم سے رابطہ کر سکتا ہے جیسا کہ: پاسٹ پیپرز، کورس سلیکشن، پروجیکٹ سلیکشن، ڈیٹ شیٹ،
فیس پے اور چینج سٹڈی پروگرام وغیرہ۔

Paid

VU LMS

اگر کوئی سٹوڈنٹ کسی مجبوری کی وجہ سے یا جاب ہو لڈر ہونے کی وجہ سے اپنا ایل ایم ایس خود ہینڈل نہیں کر سکتا تو وہ ہم سے پیڈ سروس حاصل کر
سکتا ہے کوشش کرے کے آپ ہر چیز خود اٹھپیٹ کریں اگر نہیں کر سکتے تو سروس حاصل کریں۔

Email:

waqasahmeduog@gmail.com

انشاء اللہ آپ ہماری سروس سے مطمئن ہوں گے۔

Whatsapp: 03066295623 , 03404643855

مزید اگر کسی رہنمائی کی ضرورت ہو تو وہ رابطہ کر سکتا ہے۔

Question # 2 of 10 (Start time: 08:20:43 PM, 08 June 2022)

In the following loop statement:

for i ← 1 to n

For loop executes _____ times.

Select the correct option

☐	log n
☐	n^2
☐	It is not correct statement
☒	n

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Question # 2 of 10 (Start time: 08:20:43 PM, 08 June 2022)

In the following loop statement:

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For loop executes _____ times.

Select the correct option

☐	log n
☐	n^2
☐	It is not correct statement
☒	n

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Question # 3 of 10 (Start time: 08:22:02 PM, 08 June 2022)

The main purpose of our mathematical analysis will be measuring the _____ time.

Select the correct option

☐	Compilation
☐	Debugging
☒	Execution
☐	Building

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Question # 4 of 10 (Start time: 08:22:37 PM, 08 June 2022)

Algorithm is a sequence of computational steps that _____ the input into output.

Select the correct option

☒	transform
☐	assign
☐	merge
☐	integrate

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Question # 5 of 10 (Start time: 08:22:45 PM, 08 June 2022)

The problem with the brute-force algorithm is that it uses _____ in pruning out decisions.

Select the correct option

☐	Worst-case time
☒	No intelligence
☐	Outside looping
☐	Artificial intelligence

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Question # 7 of 10 (Start time: 08:23:02 PM, 08 June 2022)

RAM is idealized machine with _____ large random-access memory.

Select the correct option

☒	Infinite
☐	Average
☐	Small
☐	Finite

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Question # 8 of 10 (Start time: 08:23:20 PM, 08 June 2022)

Asymptotic notations are used to describe _____ of an algorithm.

Select the correct option

☐	size
☐	compile time
☐	length
☒	running time

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Question # 9 of 10 (Start time: 08:23:36 PM, 08 June 2022)

Random Access Machine (RAM) can execute _____ instructions.

Select the correct option

☒	Logical and arithmetic
☐	Only logical
☐	Parallel
☐	Only arithmetic

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Question # 10 of 10 (Start time: 08:23:52 PM, 08 June 2022)

In merge sort algorithm, we split the array _____ to find index q.

Select the correct option

☐	both from start or end
☒	midway
☐	from end
☐	from start

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Question # 1 of 10 (Start time: 08:25:24 PM, 08 June 2022)

What is the best case time complexity of merge sort?

Select the correct option

☐	$O((n^2))$
☒	$O((n \log n))$
☐	$O((n \log n^2))$
☐	$O((n^2 \log n))$

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Question # 5 of 10 (Start time: 08:27:31 PM, 08 June 2022)

If we associate (x, y) integers pair to cars where x is the speed of the car and y is the negation of the price. High y value for a car means a _____ car.

Select the correct option

☐	Fast
☒	Cheap
☐	Slow
☐	Expensive

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Question # 6 of 10 (Start time: 08:28:26 PM, 08 June 2022)

Brute-force algorithm for 2D-Maxima is operated by comparing _____ pairs of points.

Select the correct option

☐	Two
☒	All
☐	Some
☐	Most

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Question # 7 of 10 (Start time: 08:28:42 PM, 08 June 2022)

In the following loop statement:

for i ← 1 to n

For loop executes _____ times.

Select the correct option

☐	n^2
☐	$\log n$
☐	It is not correct statement
☒	n

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Question # 8 of 10 (Start time: 08:28:49 PM, 08 June 2022)

Which one of the following is worst case time complexity of merge sort?

Select the correct option

☐	$O(n \log n^2)$
☐	$O((n^2 \log n))$
☐	$O(n^2)$
☒	$O(n \log n)$

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Question # 10 of 10 (Start time: 08:30:03 PM, 08 June 2022)

The asymptotic growth of $n(n+1)/2$ is:

Select the correct option

☐	$O(n \log n)$
☒	$O(n^2)$
☐	$O(n+2)$
☐	$O(n)$

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Question # 2 of 10 (Start time: 04:43:50 PM, 28 June 2022)

In _____, Leonardo of Pisa, also called Fibonacci, published a book.

Select the correct option

☐	1304
☐	1102
☒	1202
☐	1400

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Question # 3 of 10 (Start time: 04:44:33 PM, 28 June 2022)

Insertion sort is a _____ sorting algorithm.

Select the correct option

☒	In-place
☐	in-partition
☐	Unstable
☐	Not In-Place

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Question # 1 of 10 (Start time: 07:59:47 PM, 28 June 2022)

Developing a dynamic programming algorithm generally involves _____ separate steps.

Select the correct option

☐	Three
☐	One
☐	Four
☒	Two

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Question # 2 of 10 (Start time: 08:00:21 PM, 28 June 2022)

If there are duplicates in _____ sort, then each bin can be replaced by a linked list.

Select the correct option

☐	Merge
☒	Bucket
☐	Heap
☐	Quick

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Question # 3 of 10 (Start time: 08:01:18 PM, 28 June 2022)

If there are $\Theta(n^2)$ entries in edit distance matrix then the total running time is:

Select the correct option

☐	$\Theta(1)$
☐	$\Theta(n)$
☐	$\Theta(n \log n)$
☒	$\Theta(n^2)$

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Question # 6 of 10 (Start time: 08:03:47 PM, 28 June 2022)

Radix sort performs sorting the numbers _____ digit(s) at a time.

Select the correct option

☐	Two
☒	One
☐	All
☐	Three

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Question # 9 of 10 (Start time: 08:05:21 PM, 28 June 2022)

What is the worst-case time of a quick sort which happens rarely?

Select the correct option

☒	$O(n^2)$
☐	$O(\log n)$
☐	$O(n)$
☐	$O(n \log n)$

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Question # 10 of 10 (Start time: 08:06:19 PM, 28 June 2022)

Counting sort assumes that the numbers to be sorted are in the range _____.

Select the correct option

☐	k to n where k is small
☒	1 to k where k is small
☐	k to n where n is large
☐	k to n where n is small

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Question # 1 of 10 (Start time: 08:11:01 PM, 28 June 2022)

Insertion sort is a _____ sorting algorithm.

Select the correct option

☐	Unstable
☒	In-place
☐	in-partition
☐	Not In-Place

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Question # 2 of 10 (Start time: 08:11:15 PM, 28 June 2022)

_____ is the process of avoiding unnecessary repetitions by writing down the results of recursive calls and looking them up again if we need them later.

Select the correct option

☐	Function
☐	Recursion
☒	Memoization
☐	Loop

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Question # 10 of 10 (Start time: 08:16:02 PM, 28 June 2022)

As per algorithm of dynamic programing, we need to store the result(s) of _____

Select the correct option

☐	First sub-problem only
☐	Best solution only
☒	Intermediate sub-problems
☐	Final solution only

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